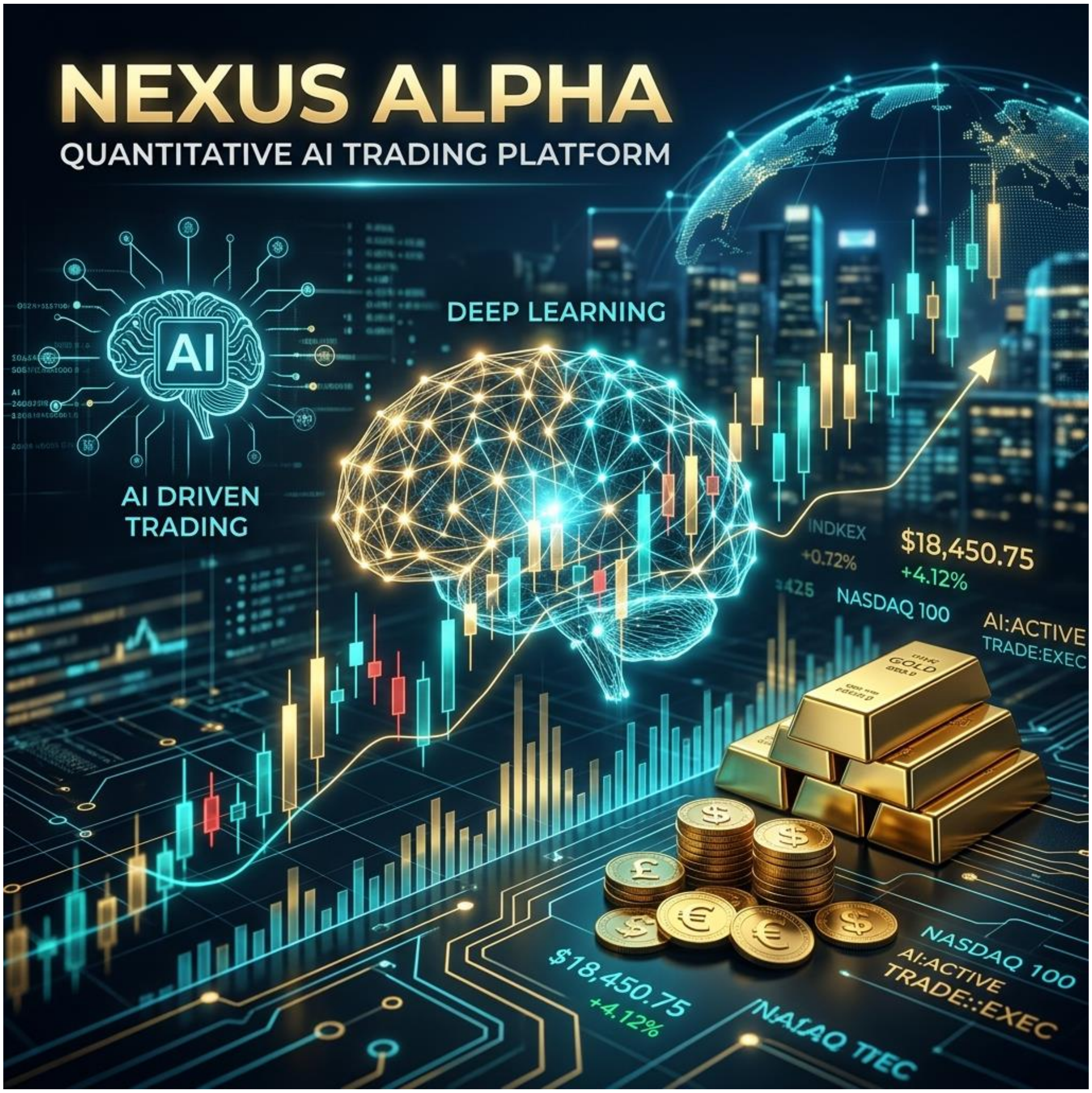
PROJECT REPORT: AI-XAU TRADER PRO



Course Code: INT428 - AI-Assisted Problem Solving
Domain: Fintech / Automated Financial Trading (XAUUSD)

Lovely Professional University, Phagwara

Project Team

- **Team Leader:** Aquib Jawaid Ansari (B.Tech CSE, Reg: 12508688, 4th Sem)
- **Team Member:** Anoop Kumar (B.Tech CSE, Reg: 12406213, 4th Sem)

1. Introduction

Financial trading, specifically in the volatile Gold (XAUUSD) market, presents a high-dimensional problem where human emotion and delayed execution often lead to losses. AI-XAU Trader Pro is an innovative AI-driven solution designed to solve the problem of market noise and emotional bias. It utilizes a sophisticated 39-Feature Ensemble Model to predict high-probability trade setups with institutional-grade accuracy.

2. Assessment Objective

- **Real-World Problem:** Solving the complexity of multi-timeframe analysis in XAUUSD trading.
- **Technical Competence:** Demonstration of API-based system design, ensemble learning, and real-time data streaming.
- **Domain Awareness:** Integration of technical indicators (RSI, EMA, ATR) and MetaTrader 5 connectivity.

3. Project Description

The project is a production-oriented trading platform that bridges Artificial Intelligence with the MetaTrader 5 (MT5) terminal. It features:

- **Multi-Timeframe Analysis:** Synchronized processing of M15, H1, and H4 data.
- **Ensemble Engine:** A hybrid architecture combining XGBoost (Gradient Boosting) and LSTM (Long Short-Term Memory) models.
- **Automated Execution:** Zero-latency trade execution with predefined Take-Profit (TP) and Stop-Loss (SL) levels.

4. Scope and Requirements

- **Domain Definition:** Specialized in XAUUSD (Gold) Spot prices.
- **Technical Stack:** Python (FastAPI), TensorFlow, Scikit-learn, MetaTrader 5 API, and Vanilla JS/CSS for the dashboard.
- **Data Handling:** Real-time ingestion of OHLC (Open, High, Low, Close) candles via MT5.

SECTION A: PROJECT OVERVIEW

- **Type of Chatbot/AI Developed:** Hybrid (Predictive ML + Rule-based Execution)
- **Platform Used for Deployment:** Web Application & Desktop Application (via MT5 Terminal)
- **Deployment Link:** Localhost Production Environment

SECTION B: MODEL & API DETAILS

- **Type of API Used:** Custom REST API (FastAPI), Local Model API (TensorFlow/XGBoost Inference), Broker API (MetaTrader 5)
- **Model Name Used:** XAU-Ensemble-v3 (39-Feature Multi-Timeframe Model)
- **Model Version:** v3.2.0 (Production Release)

SECTION C: CONTEXT & DATA HANDLING

- **Contextual Memory Usage:** Long-term memory (PostgreSQL database for trade history and model weights)

Flow of Data in the System

1. Ingestion: MT5 Terminal streams raw XAUUSD price data to the FastAPI backend.

2. Processing: signal_engine.py calculates 39 technical features (RSI, EMA Slopes, ATR) across 3 timeframes.

3. Inference: Data is passed to the XGBoost and LSTM models; a Meta-Model "Judge" calculates the final win probability.

4. UI: Real-time metrics and charts are pushed to the web dashboard via WebSockets.

5. Action: If confidence is > 95%, the trade_executor.py triggers an order in MT5.

SECTION D: MODEL CONFIGURATION & BEHAVIOR

- **Confidence Threshold:** 95% (Ensuring deterministic, high-probability trades)
- **Input Horizon:** 48 Lookback candles (for LSTM context)
- **Trade Risk:** 0.01 Lot size
- **Thinking Level:** Advanced
- **Role Assigned:** Domain Expert (Market Analyst & Professional Scalper)

SECTION E: TECHNOLOGY STACK

- **Frontend:** HTML5, CSS3 (Vanilla), Lightweight Charts API
- **Backend:** FastAPI (Python), Uvicorn
- **Database:** PostgreSQL (with SQLite Fallback)

- **AI Engine:** TensorFlow (LSTM), XGBoost, Scikit-learn
- **Integration:** MetaTrader 5 Python Library

SECTION F: IMPLEMENTATION EVIDENCE

The dashboard displays a high-tech "Neural Engine" status, a "Trade Confidence" gauge, and a real-time TradingView-style chart showing live XAUUSD price action.

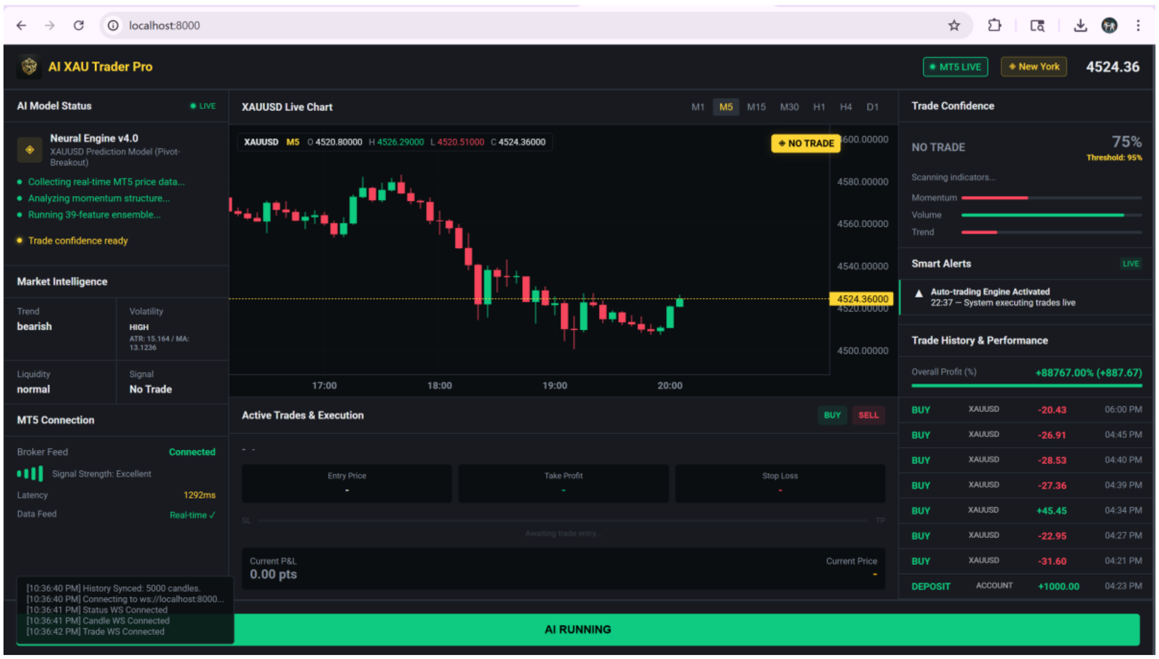


Figure 1: AI-XAU Trader Pro Dashboard in action, displaying the active Neural Engine, trade confidence metrics, and real-time live charting.

Declaration

I confirm that the AI-XAU Trader Pro is an original implementation of an ensemble-based AI system designed to solve complex financial prediction problems through innovative data engineering and multi-model consensus.

Signatures:

____

Aquib Jawaid Ansari
(Team Leader)
Date: 4 May 2026

____

Anoop Kumar
(Team Member)
Date: 4 M https:

GITHUB : https://github.com/Propopper189/Trade-Bot-Final-.git